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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER | | | | | |

CANDIDATE NUMBER | | | | |

1932759628

MATHEMATICS (SYLLABUS D) **4024/21**

Paper 2 **October/November 2012**

2 hours 30 minutes

Candidates answer on the Question Paper.

Additional Materials: Geometrical instruments
Electronic calculator

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Section A
Answer **all** questions.

Section B
Answer any **four** questions.

If working is needed for any question it must be shown in the space below that question.
Omission of essential working will result in loss of marks.
You are expected to use an electronic calculator to evaluate explicit numerical expressions.
If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.
For π, use either your calculator value or 3.142, unless the question requires the answer in terms of π.

The number of marks is given in brackets [ ] at the end of each question or part question.
The total of the marks for this paper is 100.

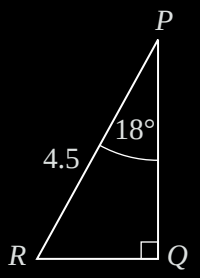
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This document consists of **23** printed pages and **1** blank page.

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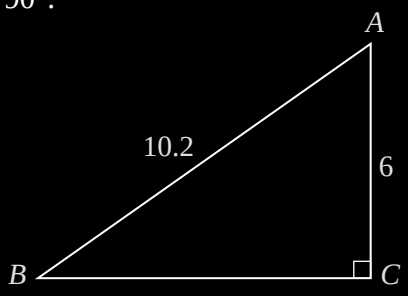
Find PQ.



Answer ..... m [2]

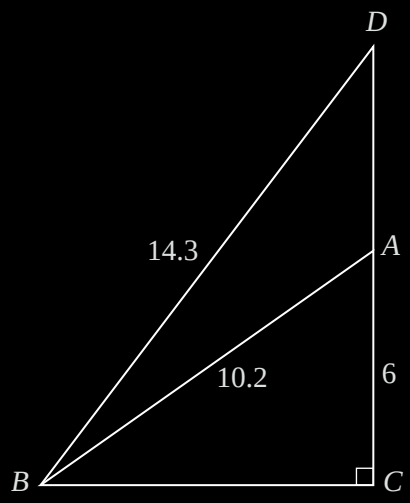
(b) In triangle ABC, $AB = 10.2$ m, $AC = 6$ m and $A\hat{C}B = 90°$.

(i) Find $A\hat{B}C$.



Answer ..... [2]

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D is the point on *CA* produced such that *BD* = 14.3 m.

Find *AD*.

Answer ..... m [4]

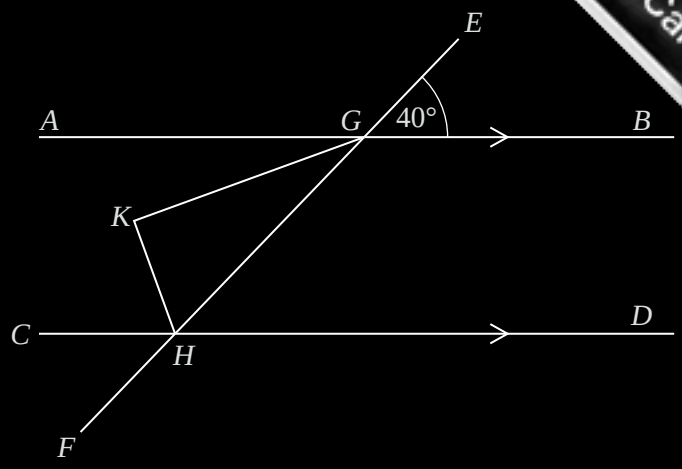
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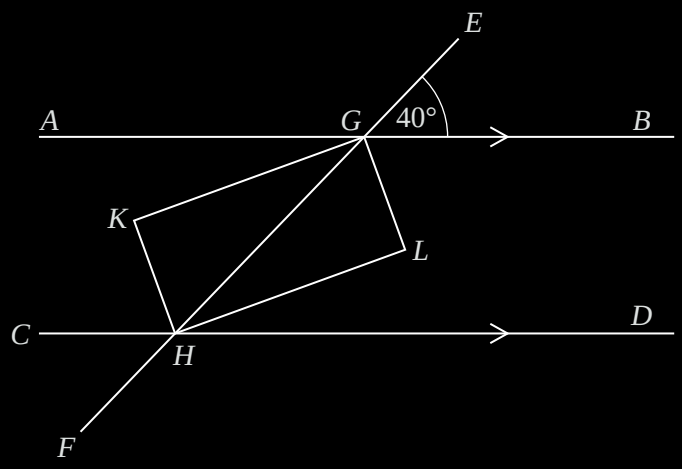
(i) Find $K\hat{G}H$.

Answer ..... [1]

(ii) Find $G\hat{H}K$.

Answer ..... [1]

(iii) The bisectors of $H\hat{G}B$ and $D\hat{H}G$ intersect at L.



State the name of the special quadrilateral *HKGL* and give your reasons.

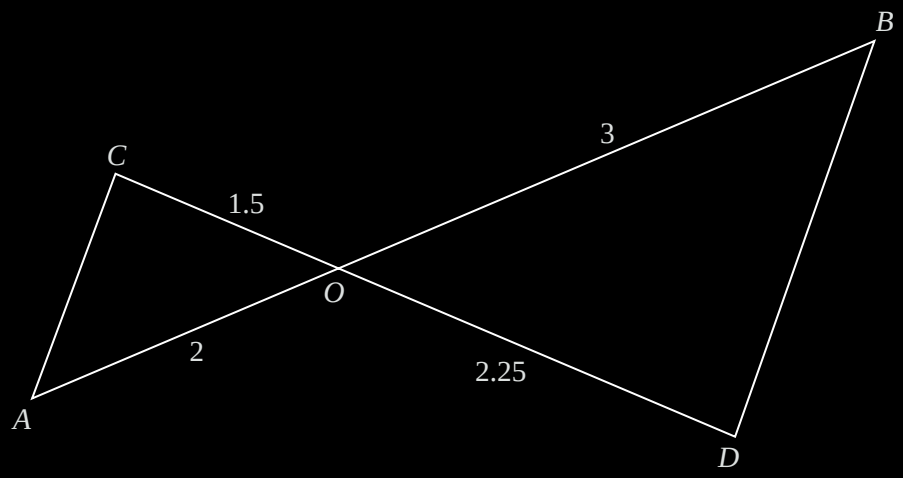
Answer HKGL is a .....

because .....

.....

.....[3]

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(i) Show that triangles *OCA* and *ODB* are similar.

Answer .....

.....

.....

.....[2]

(ii) Given that $BD = 2.7$ cm, find AC.

Answer ..... cm [1]

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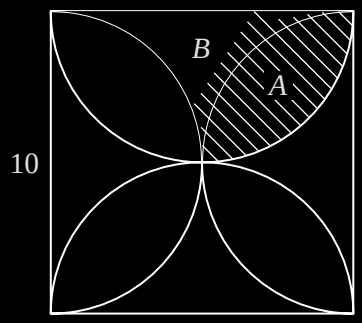


Diagram I



In **Diagram I**, four semicircles are drawn inside a square of side 10 cm.
Each semicircle has a side of the square as its diameter.
Two regions, *A* and *B*, are shown.

(a) Calculate the perimeter of region *A*.

Answer ..... cm [1]

(b) Calculate the perimeter of region *B*.

Answer ..... cm [1]

(c)

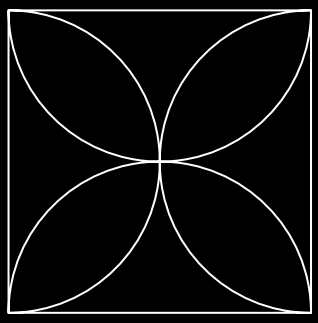


Diagram II

For **Diagram II**,

(i) draw all the lines of symmetry, [1]

(ii) state the order of rotational symmetry.

Answer ..... [1]

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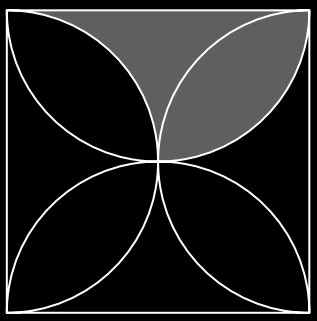


Diagram III

(i) Calculate the area of this shaded region.

Answer .....cm^2 [1]

(ii) Hence calculate the area of region *A* shown in **Diagram I**.

Answer .....cm^2 [3]

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Time (t minutes)	$60 < t \leqslant 80$	$80 < t \leqslant 90$	$90 < t \leqslant 95$	$95 < t \leqslant 100$	$100 < t \leqslant 110$	$110 < t \leqslant 130$
Number of drivers	4	10	14	20	24	8

(a) Calculate an estimate of the mean journey time.

Answer ..... minutes [3]

(b) **(i)** A driver is chosen at random.

Find the probability that the journey time for this driver is 95 minutes or less.

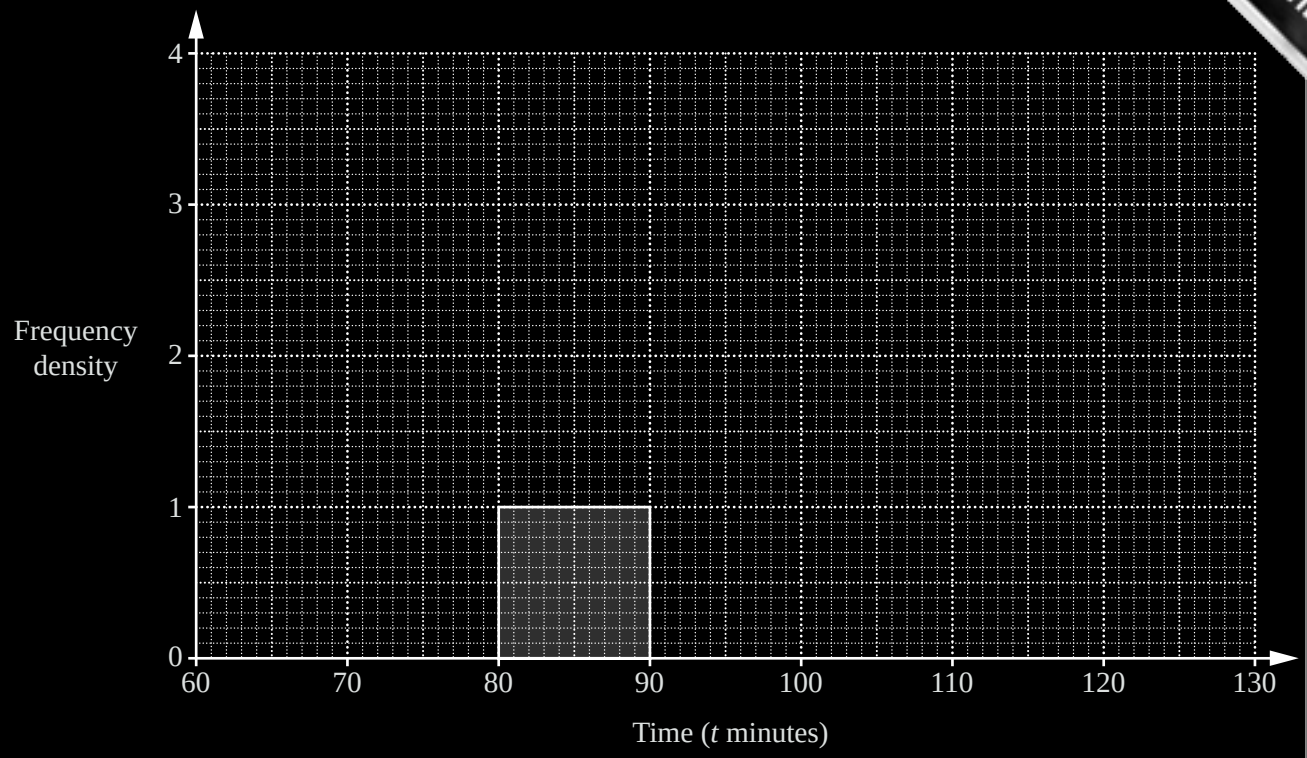
Answer .....[1]

(ii) Two drivers are chosen at random without replacement.

Calculate the probability that both their journey times are more than 100 minutes.

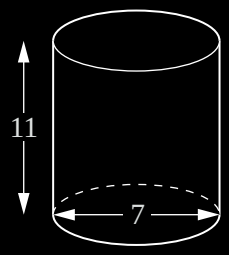
Answer .....[2]

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[3]

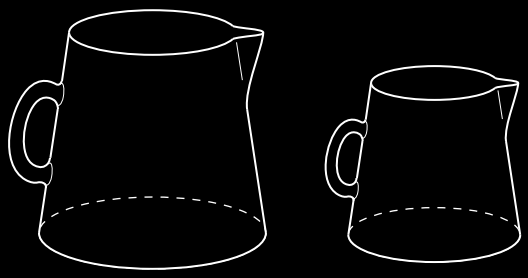
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(i) Calculate the volume of this tin.

Answer ..... cm^3 [2]

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Calculate the diameter of the base of the smaller jug.

Answer ..... cm [3]

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x	–1	–0.5	0	0.5	1	1.5	2	2.5
y	–2	0.4	1	0.6	0	–0.1	1	p

(a) Calculate p. Give your answer correct to 1 decimal place.

Answer $p =$ .....[1]

(b) On the graph paper opposite, using a scale of 2 cm to represent 1 unit on both axes,
draw a horizontal x-axis for $-2 \leqslant x \leqslant 3$ and
draw a vertical y-axis for $-3 \leqslant y \leqslant 5$.
On your axes, plot the points given in the table and join them with a smooth curve. [3]

(c) Use your graph to find all the solutions of $x^3 - 2x^2 + 1 = 0$.

Answer $x =$ .....[2]

(d) By drawing a tangent, find the gradient of the curve at the point where $x = -0.3$.

Answer .....[2]

(e) The equation $x^3 - 2x^2 - x + 2 = 0$ can be solved by finding the intersection of the graphs of $y = x^3 - 2x^2 + 1$ and the straight line $y = x + a$.

(i) Find the value of a.

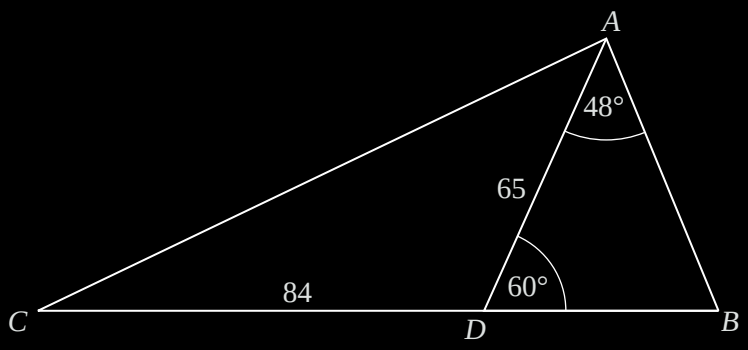
Answer $a =$ .....[1]

(ii) Hence solve the equation $x^3 - 2x^2 - x + 2 = 0$.

Answer $x =$ .....[3]

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A, B, C and *D* are four points on horizontal ground.
CDB is a straight line.
AD = 65 m and *CD* = 84 m.
$D\hat{A}B = 48°$ and $A\hat{D}B = 60°$.

(a) Calculate *AB*.

Answer .....m [3]

(b) Calculate the area of triangle *ACD*.

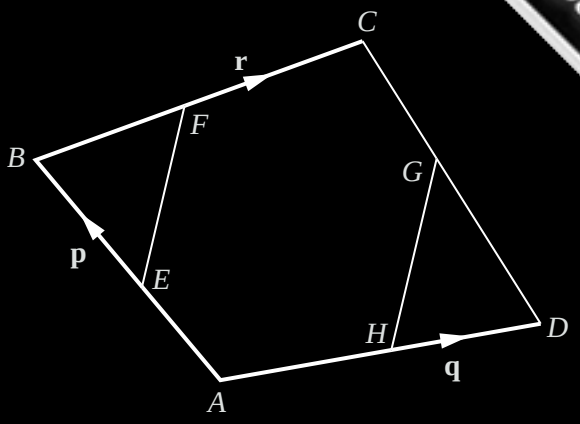
Answer ..... m^2 [2]

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(i) Find, in terms of **p**, **q** and **r** as appropriate

(a) $\overrightarrow{EF}$,

Answer .....[1]

(b) $\overrightarrow{DC}$,

Answer .....[1]

(c) $\overrightarrow{HG}$, expressing the vector as simply as possible.

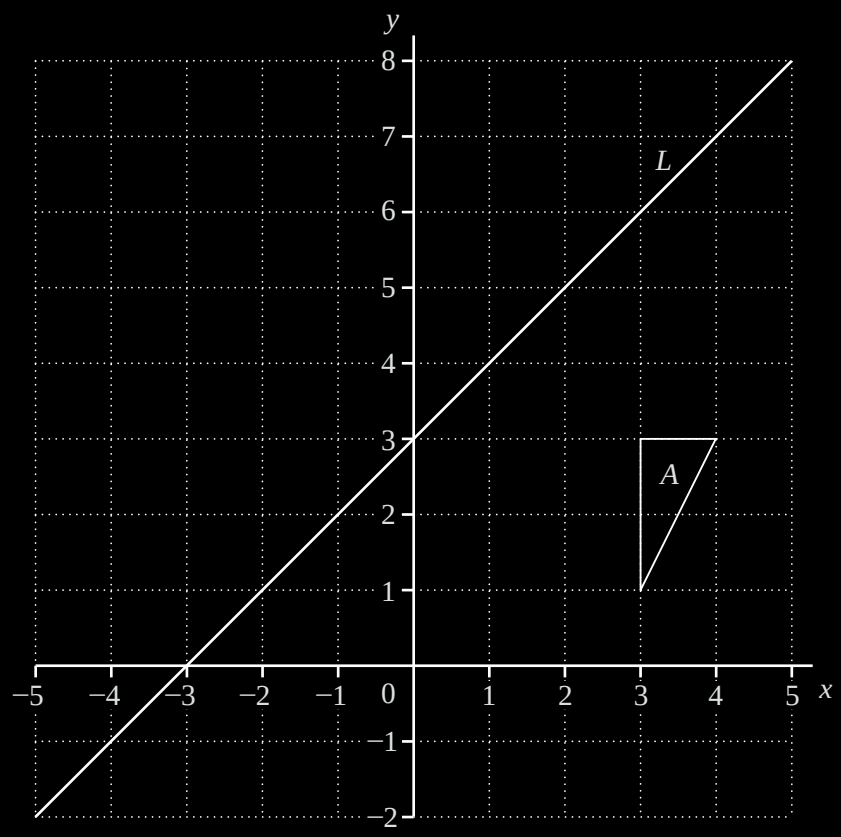
Answer .....[2]

(ii) What conclusions can be drawn about the lines *EF* and *HG*?

Answer .....

.....[1]

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The grid shows triangle *A* and line *L*.

(i) Triangle *A* is mapped onto triangle *B* by a reflection in line *L*.

Draw and label triangle *B*. [2]

(ii) Triangle *A* is mapped onto triangle *C* by a clockwise rotation of 90°, centre (0, 3).

Draw and label triangle *C*. [2]

(iii) Triangle *C* is mapped onto triangle *D* by a reflection in line *L*.

Describe the single transformation that maps triangle *B* onto triangle *D*.

Answer .....

..... [3]

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